

Fact: 1 byte = 8 bits

We talk about raw array definition here ! Or static array or traditional array (  $\bar{o}$  \_  $\bar{o}$  )

## What is array ?

- A fixed-size, contiguous memory space storing elements of the same type
- That means you cannot grow it
- There is no "insertAt" or push or pop. Doesn't mean you can't write those though

## How array work ?

Given specific index, take the width of type multiplies it by the offset.

Formular:  $a + \text{width} * \text{offset}$  when

a: address of array located in memory

- Getting at specific index, in above formula
- Insertion at specific index, basically just overwrite
- Deletion at specific index, basically just set it to 'zeroth' value

When we need bigger array, we just reallocate original array, that mean we move it to new memory address, give it bigger size and copy all contents of old array.

## How memory is interpreted in an array?

- It depends on the type of view or interpretation applied to the memory, such as an 8-bit unsigned integer view versus a 16-bit unsigned integer view, which can change how the same memory bytes are understood

## Big O ?

- ALL is  $O(1)$ , run in constant time
- It just follow that 'formula' to calculate the position then do the operation, it doesn't grow with the input